Fine-Grained Storage Simulation and I/O Performance Analysis

Technical Warm-Up Exercise – GSoC 2026, CERN HSF

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Date: *25 February 2025*

Objective

This technical warm-up exercise was conducted as part of the proposal titled *Fine-Grained Storage for the DUNE Experiment* (HSF GSoC 2026). The primary objective was to simulate simplified file storage designs and evaluate how data layout affects read and write I/O performance. The experiment specifically aimed to compare coarse-grained storage (e.g., single or chunked files) versus fine-grained storage (individual files) in terms of access efficiency, and to understand the trade-offs involved.

Methodology

A dataset of 100,000 sub-records was generated, each with a size randomly distributed between 1 KB and 2 KB. Data content consisted of random bytes, and a fixed random seed ensured reproducibility. The following three storage strategies were implemented in Python:

1. Single Large File: All sub-records were written sequentially into a single binary file. An index containing byte offsets and sizes for each record enabled fast random access.

2. Chunked Files: Every 1,000 sub-records were grouped into a separate file. A per-file index maintained internal positions to allow random access.

3. Individual Files: Each sub-record was stored as its own file with a deterministic filename based on its index.

Two access patterns were benchmarked for each strategy: *Sequential Read*, where all records were read in order, and *Random Read*, where 1000 records were randomly selected and accessed. All read and write operations were timed, and both total duration and average latency per record were calculated.

Results

Strategy	Write Time	Sequential Read	Random Read (1000)
Single Large File	0.453 s	0.143 s	0.007 s
Chunked Files	0.410 s	0.143 s	0.098 s
Individual Files	4.988 s	3.628 s	0.036 s

Insights and Learnings

The single large file approach achieved superior performance, especially for random reads, due to minimal system overhead and direct indexed access. Chunked files exhibited competitive performance, particularly during writing, thanks to fewer open/close operations. Individual files suffered the most from filesystem overhead, demonstrating significantly higher write and sequential read times. One key takeaway was that storage layout and access pattern alignment play a vital role in optimizing I/O efficiency.

Through this exercise, I learned how file system operations (e.g., opening, seeking, reading) compound at scale. I also gained hands-on experience in designing and benchmarking efficient indexing structures and discovered the hidden performance costs of fine-grained storage strategies. Additionally, I saw firsthand how simple benchmarking tools can expose meaningful architectural insights.

Conclusion

This experiment confirmed that coarse-grained storage strategies outperform fine-grained ones in both sequential and random access contexts. While individual files may offer modularity, the associated overhead makes them unsuitable for large-scale workloads such as DUNE. Chunked or indexed single-file designs strike an effective balance between granularity and performance.